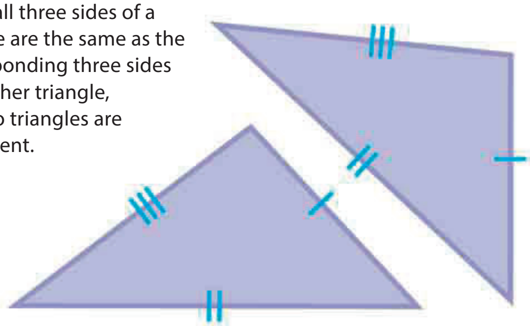
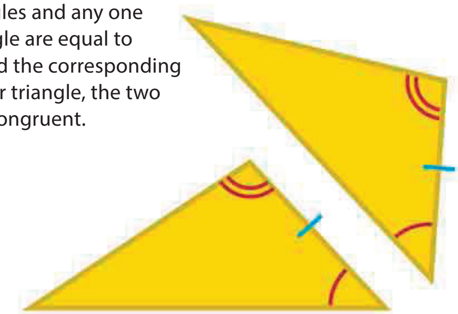
How to tell if triangles are congruent

It is possible to tell if two triangles are congruent without knowing the lengths of all of the sides or the sizes of all of the angles—knowing just three measurements will do. There are four groups of measurements.

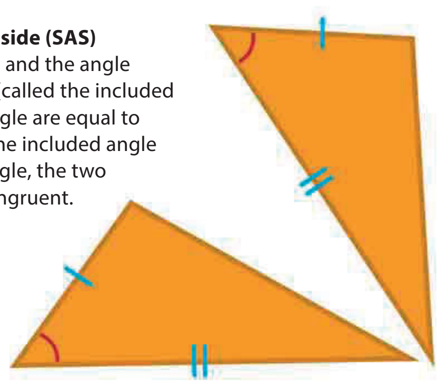
▷ **Side, side, side (SSS)**
When all three sides of a triangle are the same as the corresponding three sides of another triangle, the two triangles are congruent.



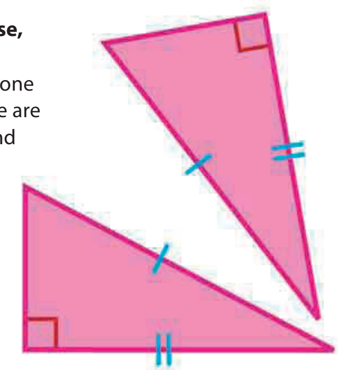
▷ **Angle, angle, side (AAS)**
When two angles and any one side of a triangle are equal to two angles and the corresponding side of another triangle, the two triangles are congruent.



▷ **Side, angle, side (SAS)**
When two sides and the angle between them (called the included angle) of a triangle are equal to two sides and the included angle of another triangle, the two triangles are congruent.

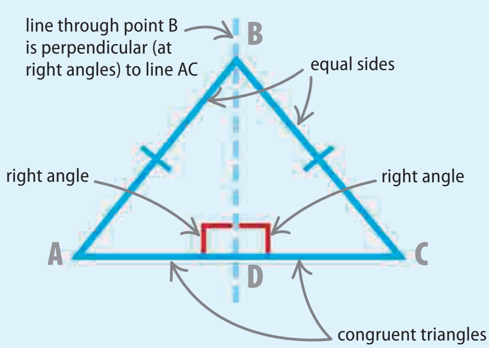


▷ **Right angle, hypotenuse, side (RHS)**
When the hypotenuse and one other side of a right triangle are equal to the hypotenuse and one side of another right triangle, the two triangles are congruent.

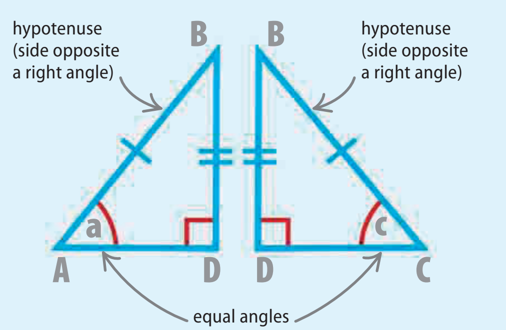


Proving an isosceles triangle has two equal angles

An isosceles triangle has two equal sides. Drawing a perpendicular line helps prove that it has two equal angles too.



Draw a line perpendicular (at right angles) to the base of an isosceles triangle. This creates two new right triangles. They are congruent—identical in every way.



The perpendicular line is common to both triangles. The two triangles have equal hypotenuses, another pair of equal sides, and right angles. The triangles are congruent (RHS) so angles "a" and "c" are equal.